

WIN
SERVER

AZ-800

Administering Windows Server Hybrid Core Infrastructure

STUDY NOTES

Complete Windows Server Administration Reference Guide

2025 EDITION

10

Sections

40+

Topics

200+

Commands

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Microsoft Windows Server Certification Preparation Guide

AZ-800 Study Notes | 2025 Edition

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10 Sections

40+ Topics

200+ Commands

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PREFACE & INTRODUCTION

Welcome to the AZ-800 Windows Server Study Notes — your concise, structured reference guide for rapid revision and exam preparation for the Administering Windows Server Hybrid Core Infrastructure certification.

Unlike the Q&A format, this guide presents each topic with key concepts, important points to remember, and a command reference table — making it perfect for last-minute revision, building mental models, or quick on-the-job lookup.

How to use this guide: Each section covers a major AZ-800 exam domain. Under each topic heading you will find bullet-point key concepts with bold highlights for exam-critical items, followed by a command reference table with practical examples. Pay special attention to bold terms — these are the most frequently tested concepts.

Topics covered: Windows Server Foundations • Active Directory Domain Services • DNS & DHCP • Hyper-V & Virtualization • Storage & File Services • Network Policy & Remote Access • Windows Server Security • High Availability & Clustering • Monitoring & Troubleshooting • Hybrid & Cloud Integration

Best of luck in your AZ-800 certification journey! Practice the commands in a lab environment — the exam tests hands-on knowledge.

— *Akhilesh Chaudhari*

Section 1: Windows Server Foundations

Server Core, PowerShell, GPO, Roles & Features

Windows Server 2022 Editions & Installation

- Datacenter: unlimited VMs, all features — SDN, S2D, Shielded VMs. Standard: 2 VMs per license. Essentials: small business ≤25 users
- Server Core (recommended): no GUI — smaller attack surface, fewer updates, lower footprint — managed via PowerShell, WAC, remote Server Manager
- Server with Desktop Experience: full GUI — easier but higher attack surface and update frequency
- Switch Core↔Desktop: requires reinstall. Core is default in most deployment guides
- Nano Server: container base image only (no longer installable as OS) — smallest footprint

Get-WindowsFeature	List all roles/features and their install status
Install-WindowsFeature -Name DNS -IncludeManagementTools	Install DNS role with tools
Remove-WindowsFeature -Name Telnet-Client	Remove a feature
sconfig.cmd	Server Core text-based configuration menu
Server Manager □ Add Roles and Features	GUI wizard for role/feature installation

PowerShell Essentials for Server Admin

- Cmdlet format: Verb-Noun — Get-Service, Set-Item, New-ADUser, Remove-Item
- Pipeline: pass objects between cmdlets — Get-Process | Where CPU -gt 50 | Stop-Process
- PowerShell Remoting: runs commands on remote servers via WinRM (port 5985 HTTP / 5986 HTTPS)
- Enable remoting: Enable-PSRemoting. Interactive session: Enter-PSSession -ComputerName server1
- Run on multiple servers: Invoke-Command -ComputerName s1,s2 -ScriptBlock {Get-Service}
- Modules: import with Import-Module; auto-loaded when cmdlet used. Find: Find-Module in PSGallery

Enter-PSSession -ComputerName server1	Interactive remote PowerShell session
Invoke-Command -ComputerName s1,s2 {Get-Service}	Run on multiple remote servers
Get-Command -Module ActiveDirectory	List all AD module cmdlets
Get-Help Get-ADUser -Examples	Show cmdlet usage examples
Set-ExecutionPolicy RemoteSigned	Allow local/signed remote scripts to run
\$cred = Get-Credential; New-PSSession -Credential \$cred	Remote session with credentials

Group Policy (GPO) – Core Concepts

- GPO processing order (LSDOU): Local □ Site □ Domain □ OU (parent to child OU). Last applied wins for conflicts
- Computer Configuration: applied at startup, affects machine regardless of user. Refreshed every 90 min
- User Configuration: applied at logon, follows user to any machine. Refreshed every 90 min
- Enforced (No Override): parent GPO cannot be blocked by child OUs – use sparingly
- Block Inheritance: OU blocks all parent GPOs – overridden by Enforced policies
- WMI Filters: apply GPO only to computers matching WMI query (e.g., only Windows 11 machines)
- Security Filtering: apply GPO only to specific users/groups/computers (default: Authenticated Users)

<code>gpupdate /force</code>	<i>Force immediate GPO refresh on local machine</i>
<code>gpresult /r</code>	<i>Show RSoP (Resultant Set of Policy) – applied GPOs for current user/computer</i>
<code>gpresult /h gpreport.html</code>	<i>Generate detailed HTML GPO report</i>
<code>Get-GPO -All</code>	<i>List all GPOs in domain</i>
<code>New-GPO -Name 'Security Baseline' New-GPLink -Target 'OU=Servers,DC=corp,DC=com'</code>	<i>Create + link GPO</i>
<code>Invoke-GPUUpdate -Computer server1 -Force</code>	<i>Remote GPO update via PowerShell</i>

Windows Admin Center & Server Management

- WAC: free browser-based management – replaces many MMC snap-ins. Install as gateway for multi-server management
- WAC manages: servers, clusters, HCI (S2D), Azure Stack HCI, Azure Arc-enrolled servers
- RSAT (Remote Server Administration Tools): management tools for client machines to manage servers
- WSUS: Windows Server Update Services – centralize patch management. Clients point to WSUS via Group Policy
- WSUS approval workflow: sync from Microsoft □ approve for computer groups □ deploy
- DSC (Desired State Configuration): declare desired server state in scripts – LCM enforces and corrects drift

<code>winrm quickconfig</code>	<i>Enable WinRM for remote management (required for WAC/PSRemoting)</i>
<code>Install-WindowsFeature -Name RSAT-AD-Tools</code>	<i>Install AD management tools on server</i>
<code>Start-DscConfiguration -Path .\config -Wait -Verbose</code>	<i>Apply DSC configuration</i>
<code>Get-DscLocalConfigurationManager</code>	<i>Show LCM settings (refresh mode, config mode)</i>
<code>Test-DscConfiguration</code>	<i>Check if current state matches desired state</i>

Section 2: Active Directory Domain Services

DCs, FSMO, Sites, Replication, Trusts & Authentication

AD DS Architecture

- Forest: security boundary — shares schema, configuration, GC. All domains trust each other transitively
- Domain: administrative partition — owns OUs, policies, domain-wide settings. Multiple domains possible in one forest
- Domain Controller: stores NTDS.DIT (AD database), handles Kerberos auth. Min 2 DCs per domain for redundancy
- Global Catalog (GC): partial read-only copy of all forest objects — required for UPN logon, universal group membership, Exchange
- OU: container for delegation + GPO application — NOT a security boundary. Design based on admin requirements
- Functional Levels: minimum DC OS version. Raising is irreversible. 2012R2 DFL enables AD Recycle Bin

Get-ADDomain	View domain info including functional level, PDC emulator
Get-ADForest	View forest info including schema master, domain naming master
Get-ADDomainController -Filter *	List all DCs in domain
netdom query fsmo	Identify all 5 FSMO role holders
Move-ADDirectoryServerOperationMasterRole -OperationMasterRole PDCEmulator -Identity DC2	Transfer FSMO role
dcpromo /unattend	Legacy DC promotion (modern: Install-ADDSDomainController)

FSMO Roles — Quick Reference

- Schema Master (forest-wide, 1): controls AD schema changes — add new attributes/classes
- Domain Naming Master (forest-wide, 1): add/remove domains from forest
- PDC Emulator (domain-wide, 1 per domain): time sync authority, password change propagation, account lockout processing, GPO editing point — *most critical FSMO role*
- RID Master (domain-wide, 1 per domain): issues pools of Relative IDs to DCs for creating new AD objects
- Infrastructure Master (domain-wide, 1 per domain): updates cross-domain object references. Should NOT be on GC server (unless all DCs are GCs)
- Seize roles (emergency) with ntdsutil when original DC cannot be recovered

netdom query fsmo	Quick FSMO role holder lookup
Get-ADDomain Select PDCEmulator, RIDMaster, InfrastructureMaster	PowerShell FSMO query

Get-ADForest Select SchemaMaster, DomainNamingMaster	Forest FSMO roles
ntdsutil 'roles' 'connections' 'connect to server DC2' 'q' 'seize PDC' 'q' 'q'	Seize PDC Emulator
w32tm /query /status	Check time sync status (PDC Emulator is authoritative source)

AD Replication & Sites

- Intra-site replication: immediate (15 sec notification), uses RPC over IP — DCs in same site
- Inter-site replication: scheduled via site links (default 180 min), uses ISTG-determined topology
- KCC (Knowledge Consistency Checker): auto-builds replication topology (connection objects) — runs every 15 min
- Sites: physical locations — control replication schedule + authentication locality (clients use closest DC)
- Site Links: define cost (lower = preferred), schedule, and replication interval between sites
- USN (Update Sequence Number): tracks changes per DC — used for replication conflict resolution

repadmin /showrepl	Show replication status for all DCs
repadmin /replsummary	Summary of replication health across all DCs
repadmin /syncall /AdeP	Force sync of all directory partitions
dcdiag /test:replications	Comprehensive replication health test
repadmin /kcc	Force KCC to recalculate replication topology
repadmin /showrepl * /errorsonly	Show only replication errors

Kerberos, NTLM & Authentication

- Kerberos: default AD auth protocol — client gets TGT from KDC, then service tickets. Requires time sync ± 5 min
- krbtgt account: key for all Kerberos tickets — reset twice after DC compromise (Golden Ticket mitigation)
- SPN (Service Principal Name): uniquely identifies service for Kerberos — duplicate SPNs cause auth failures
- NTLM: legacy challenge-response — still used for workgroups, IP auth, legacy apps. Vulnerable to pass-the-hash
- Protected Users group: blocks NTLM, DES/RC4 Kerberos, credential caching, delegation — adds significant auth protection
- Test with klist (Kerberos tickets), nltest /sc_verify (secure channel), event IDs 4624/4625/4771

klist tickets	Show current user's Kerberos ticket cache
klist purge	Clear Kerberos ticket cache (force re-authentication)
setspn -L servername	List SPNs registered for a service account or computer
setspn -X	Find duplicate SPNs in domain (critical for troubleshooting Kerberos)

nltest /sc_verify:domain.com	<i>Verify secure channel between member and DC</i>
nltest /dsgetdc:domain.com	<i>Find a domain controller for authentication</i>

Section 3: DNS & DHCP

DNS Zones, Records, DHCP Scopes, Failover & Integration

DNS Server – Zone Types & Records

- AD-Integrated zones: stored in AD database — secure dynamic updates, multi-master replication, no zone transfer needed
- Primary zone: authoritative read-write (file-based or AD-integrated). Secondary: read-only replica. Stub: NS records only
- Critical AD DNS records: `_ldap._tcp.dc._msdcs` (DC locator), `_kerberos._tcp` (Kerberos), `_msdcs` zone (GC, PDC)
- Key record types: A (hostname→IPv4), AAAA (IPv6), CNAME (alias), MX (mail), SRV (services), PTR (reverse), TXT (SPF/DKIM)
- Scavenging: auto-remove stale dynamic records. No-Refresh (7d) + Refresh (7d) intervals — enable on zone AND server

<code>nslookup -type=SRV _ldap._tcp.dc._msdcs.domain.com</code>	Verify AD DC locator SRV records exist
<code>dcdiag /test:dns</code>	Comprehensive AD DNS validation test
<code>Resolve-DnsName -Name server1 -Type A</code>	PowerShell DNS query
<code>dnscmd /scavengesserver</code>	Trigger DNS scavenging immediately
<code>ipconfig /registerdns</code>	Force client to re-register DNS records
<code>Get-DnsServerZone</code>	List all DNS zones on server

DNS Forwarding & Conditional Forwarding

- Forwarders: DNS queries for unknown zones forwarded to specific DNS servers (e.g., ISP, 8.8.8.8) instead of root hints
- Conditional Forwarders: forward queries for specific domain names to specific servers — used for inter-forest resolution
- Root hints: fallback when forwarders fail — list of internet root DNS servers. Disable for internal-only DNS
- Split-brain DNS: same zone name with different records for internal (private IPs) vs external (public IPs) clients
- DNS over HTTPS (DoH): Windows Server 2022 supports encrypted DNS queries — configure in DNS client settings

<code>Add-DnsServerConditionalForwarderZone -Name 'partner.com' -MasterServers 10.1.1.1</code>	Add conditional forwarder
<code>Set-DnsServerForwarder -IPAddress 8.8.8.8, 1.1.1.1</code>	Set forwarders
<code>dnscmd /info /forwarders</code>	View current forwarder configuration
<code>Get-DnsServerRootHint</code>	View root hint servers

DHCP Server — Scopes, Options & Failover

- Scope: IP address range for a subnet. Must be activated. Includes exclusions, options, lease duration (default 8 days)
- DHCP Options: 003=Router/Gateway, 006=DNS Servers, 015=DNS Domain, 066/067=PXE boot (TFTP server/filename)
- Reservations: always give same IP to specific MAC address. Better than static IP — still managed by DHCP
- DHCP Failover (2012+): Hot Standby (active+passive, 5% reserve for standby) or Load Sharing (50/50 split, both active)
- DHCP Authorization: Windows DHCP servers must be authorized in AD to prevent rogue servers
- Superscope: multiple scopes serving same physical subnet (multi-net). Multicast scope: Class D addresses (224-239.x.x.x)

Get-DhcpServerv4Scope -ComputerName dhcpsvr	List all DHCP scopes
Get-DhcpServerv4Lease -ScopeId 192.168.1.0 -ComputerName dhcpsvr	Show active leases
Add-DhcpServerv4Failover -Name 'ScopeFailover' -ScopeId 192.168.1.0 -PartnerServer DHCP2 -Mode HotStandby	Configure failover
Add-DhcpServerInDC -DnsName dhcpsvr.corp.com	Authorize DHCP server in AD
Export-DhcpServer -File C:\dhcp-backup.xml	Export DHCP configuration
ipconfig /release && ipconfig /renew	Release and renew DHCP lease

Section 4: Hyper-V & Virtualization

VM Management, Networking, Storage, Replica & Advanced Features

Hyper-V Core Concepts

- Hyper-V is a Type 1 bare-metal hypervisor — runs directly on hardware, management OS is privileged root partition
- Gen 1: BIOS, IDE, legacy NIC — max compatibility. Gen 2: UEFI, SCSI boot, Secure Boot, faster — use for supported OSes
- VHD: max 2 TB, legacy. VHDX: max 64 TB, corruption-resistant, 4K sector support — always use VHDX for new VMs
- Disk types: Fixed (best performance), Dynamic (space-efficient, slower), Differencing (child shares parent — VDI/test)
- Integration Services: Heartbeat, Time Sync, Data Exchange, VSS Backup, Shutdown, Guest Services — must be current in guest
- Dynamic Memory: auto-adjust RAM. Set: Startup/Minimum/Maximum RAM + Buffer % + Memory Weight for priority

```
New-VM -Name VM1 -Generation 2  
-MemoryStartupBytes 4GB -VHDPATH  
C:\VMs\VM1.vhdx
```

Create new Gen2 VM

```
Set-VMMemory VM1  
-DynamicMemoryEnabled $true  
-MinimumBytes 1GB -MaximumBytes 8GB
```

Configure dynamic memory

```
Get-VM | Select Name, State, CPUUsage,  
MemoryAssigned
```

Quick VM status overview

```
Checkpoint-VM -Name VM1 -SnapshotType  
Production
```

Create production (VSS) checkpoint

```
Convert-VHD -Path old.vhd  
-DestinationPath new.vhdx -VHDType  
Dynamic
```

Convert VHD to VHDX

Hyper-V Networking

- External switch: VMs + host access physical network. Internal: VMs + host only (no physical). Private: VMs only (isolated)
- SET (Switch Embedded Teaming): NIC teaming built into vSwitch — supports RDMA and SR-IOV. Replaces deprecated LBFO for Hyper-V
- SR-IOV: NIC presents virtual functions directly to VMs — near-native performance, but prevents live migration
- RDMA/SMB Direct: zero-copy network I/O for server-to-server storage (S2D, SOFS). Requires RDMA-capable NICs (RoCE/iWARP)
- VLAN tagging per VM adapter: Set-VMNetworkAdapterVlan -VMName VM1 -Access -VlanId 100

New-VMSwitch -Name 'External' -NetAdapterName 'Ethernet' -AllowManagementOS \$true	Create external vSwitch
New-VMSwitch -Name 'SET-Team' -NetAdapterName 'NIC1','NIC2' -EnableEmbeddedTeaming \$true	Create SET team vSwitch
Set-VMNetworkAdapterVlan -VMName VM1 -Access -VlanId 100	Assign VLAN to VM NIC
Enable-NetAdapterRdma -Name 'NIC1'	Enable RDMA on network adapter
Get-VMNetworkAdapter -VMName VM1	Show VM network adapter settings

Live Migration, Replica & Checkpoints

- Live Migration: move running VM between hosts with no downtime. Requires: compatible CPUs, shared/CSV or SMB storage
- Shared Nothing Live Migration: migrates both VM memory AND storage — no shared storage needed
- Hyper-V Replica: async replication to DR site. Replication every 30s/5min/15min. HTTP/HTTPS. No shared storage needed
- Replica failover types: Planned (graceful, zero loss), Test (non-disruptive, creates test VM), Unplanned (disaster, possible loss)
- Production checkpoints (VSS-based): application-consistent — USE for production. Standard checkpoints: captures memory state — NOT for production
- Disable automatic checkpoints in production: Set-VM -Name VM1 -AutomaticCheckpointsEnabled \$false

Move-VM -Name VM1 -DestinationHost Host2 -IncludeStorage -DestinationStoragePath C:\VMs	Shared Nothing Live Migration
Enable-VMReplication -VMName VM1 -ReplicaServerName ReplicaHost -AuthenticationType Certificate	Enable Hyper-V Replica
Start-VMFailover -VMName VM1 -ComputerName ReplicaHost -Prepare	Planned failover
Start-VMInitialReplication -VMName VM1	Start initial replication
Get-VMReplication	Check replication status and health
Set-VM -Name VM1 -CheckpointType Production -AutomaticCheckpointsEnabled \$false	Configure production checkpoints

Shielded VMs & Security

- Shielded VMs: protect against compromised fabric/host admins — vTPM, BitLocker inside VM, HGS-controlled key release

- HGS (Host Guardian Service): validates host integrity before releasing VM keys. Modes: Admin-trusted (AD group) or TPM-trusted (more secure)
- Nested Virtualization: run Hyper-V inside a VM — enable: Set-VMProcessor -ExposeVirtualizationExtensions \$true
- Azure Stack HCI: Hyper-V + S2D + SDN on validated hardware. Annual subscription, Azure Arc-integrated, continuously updated
- VM resource metering: Enable-VMResourceMetering / Measure-VM — track CPU/memory/network per VM for chargeback

Set-VMProcessor -VMName VM1 -ExposeVirtualizationExtensions \$true	Enable nested virtualization
Get-VMSecurity VM1	Check Shielded VM status and vTPM
Enable-VMResourceMetering -VMName VM1	Enable resource metering for chargeback
Measure-VM -Name VM1	Get current resource usage metrics
Set-VMProcessor VM1 -Count 4 -Maximum 50 -Reserve 10	Configure vCPU count and limits

Section 5: Storage & File Services

Storage Spaces Direct, SMB, DFS, NTFS, iSCSI & Dedup

Storage Spaces & Storage Spaces Direct

- Storage Spaces: pool physical disks □ virtual disks with resiliency. Resiliency: Simple (none), Mirror (2-way/3-way), Parity
- Storage Spaces Direct (S2D): extends Storage Spaces across cluster nodes — replaces SAN. Uses CSV + SMB Direct (RDMA)
- S2D minimum: 2 nodes all-flash, 3 nodes hybrid. Maximum 16 nodes. Requires Datacenter edition
- Storage Tiering: hot data □ SSD, cold data □ HDD automatically. Real-time tier = SSD write cache for write bursts
- NTFS: supports ACLs, compression, EFS, quotas, hard links. ReFS: no compression/EFS but integrity streams, block cloning, large volume support
- Deduplication: remove duplicate data blocks. File server: 50-70% savings. VDI: 80-95% savings. Not for OS volumes

Enable-ClusterS2D -CacheMode Enabled -AutoConfig:0 -SkipEligibilityChecks	Enable Storage Spaces Direct
New-StoragePool -FriendlyName DataPool -StorageSubSystemFriendlyName * -PhysicalDisks (Get-PhysicalDisk -CanPool \$true)	Create storage pool
New-VirtualDisk -StoragePoolFriendlyName DataPool -FriendlyName DataVD -ResiliencySettingName Mirror -Size 1TB	Create mirrored virtual disk
Enable-DedupVolume -Volume D: -UsageType Default	Enable deduplication on volume
Get-DedupStatus	Check deduplication savings and status
Get-StoragePool Get-VirtualDisk Get-Disk Get-Partition Get-Volume	Full storage hierarchy view

SMB & File Sharing

- SMBv1: DISABLE immediately — vulnerable (WannaCry/EternalBlue). SMBv3: encryption, multichannel, transparent failover
- SMB Multichannel: automatic aggregation of multiple NICs for higher throughput + fault tolerance (no config needed if hardware supports)
- SMB over QUIC (Server 2022 Azure Edition): SMB over UDP 443 — access file shares over internet without VPN
- Share permissions vs NTFS: effective access = most restrictive of both. Best practice: Share=Everyone Full Control, control via NTFS only

- Scale-Out File Server (SOFS): active-active clustered file server — all nodes serve SMB simultaneously. Requires CSV

Set-SmbServerConfiguration -EnableSMB1Protocol \$false -Force	Disable SMBv1 immediately
Get-SmbServerConfiguration Select EnableSMB1Protocol, EnableSMB2Protocol, EncryptData	Check SMB configuration
New-SmbShare -Name 'Data' -Path 'D:\Data' -FullAccess 'CORP\Admins' -ChangeAccess 'CORP\Users'	Create SMB share with permissions
Get-SmbMultichannelConnection	Verify SMB multichannel is active
Set-SmbServerConfiguration -EncryptData \$true	Enforce SMB encryption for all connections

DFS, iSCSI & Other Storage

- DFS Namespaces: unified path (\\domain\dfs\) aggregating shares from multiple servers — clients see one path
- DFS Replication (DFS-R): multi-master file replication using RDC (only changed blocks). Replaced FRS — replicate SYSVOL
- iSCSI: block storage over TCP/IP. Windows role: iSCSI Target Server (exports VHDs as targets) / iSCSI Initiator (connects)
- MPIO: multiple paths to storage for fault tolerance + load balancing. Required for enterprise iSCSI/FC SAN
- Storage Replica (Datacenter): synchronous or async block-level replication. Synchronous = zero data loss
- FSRM: quota management, file screening (block .mp3/.exe), storage reports, file classification

New-DfsnRoot -TargetPath '\\server\share' -Type DomainV2 -Path '\\domain\dfs'	Create DFS namespace
New-DfsReplicationGroup -GroupName 'DataSync' Add-DfsrMember -ComputerName server1,server2	Create DFS-R group
iscsicpl.exe	Open iSCSI Initiator configuration GUI
New-IscsiVirtualDisk -Path D:\iSCSI\disk1.vhdx -SizeBytes 500GB	Create iSCSI virtual disk target
New-SRPartnership -SourceComputerName SRV1 -SourceRGName RG1 -SourceVolumeName D: -DestinationComputerName SRV2 -DestinationRGName RG2 -DestinationVolumeName D:	Configure Storage Replica

Section 6: Network Policy & Remote Access

NPS/RADIUS, VPN, DirectAccess, AOVPN & RDS

NPS (RADIUS) & 802.1X

- NPS (Network Policy Server): Microsoft RADIUS server — centralized auth for VPN, wireless, wired 802.1X, RD Gateway
- Network Policies: conditions (day/time, user group, NAS type) □ allow/deny + settings (VLAN via VSA)
- 802.1X: port-based NAC. Supplicant (client) □ Authenticator (switch/AP) □ Authentication Server (NPS). Uses EAP
- EAP methods: PEAP-MSCHAPv2 (username/password + server cert), EAP-TLS (client + server certs — most secure)
- RADIUS clients: VPN servers, wireless APs, managed switches — must register IP and shared secret in NPS

Get-NpsRadiusClient	List all configured RADIUS clients
Export-NpsConfiguration -Path C:\nps-backup.xml	Backup NPS configuration
Import-NpsConfiguration -Path C:\nps-backup.xml	Restore NPS configuration
netsh nps show config	Show NPS configuration summary
Get-WinEvent -LogName 'Security' -FilterHashtable @{Id=6272,6273}	View NPS grant/deny events

VPN Protocols & Always-On VPN

- IKEv2: modern, fast reconnect (MOBIKE), strong crypto — preferred for AOVPN user tunnel and mobile devices
- SSTP: TLS over TCP 443 — works through firewalls blocking UDP. Windows-only native client
- L2TP/IPsec: widely compatible, requires cert or pre-shared key. UDP 500/4500
- PPTP: deprecated and insecure — never use in new deployments
- Always-On VPN (AOVPN): modern DirectAccess replacement. Device tunnel (machine auth) + User tunnel (user auth). Supports non-domain clients
- Split vs Full Tunnel: Split = only corporate traffic via VPN (better performance). Full = all traffic via VPN (better security inspection)

Add-VpnConnection -Name 'CorpVPN' -ServerAddress vpn.corp.com -TunnelType IKEv2 -AuthenticationMethod MachineCertificate	Create IKEv2 VPN connection
Get-VpnConnection	List VPN connections on machine
netsh ras show activeusers	Show currently connected VPN users on RRAS server
Get-RemoteAccessConnectionStatistics	Show VPN connection stats

Remote Desktop Services (RDS)

- RD Session Host (RDSH): multiple users share one server — each gets isolated Windows session
- RD Connection Broker: load balances + reconnects users to existing sessions. Requires SQL Server for HA
- RD Gateway: wraps RDP in HTTPS (port 443) — external access without VPN. Integrates with NPS for auth
- RD Licensing: manages RDS CALs (per User or per Device). 120-day grace period then connections blocked
- RemoteApp: publish specific applications rather than full desktop — accessed from Start menu or web portal
- RDS CAL types: Per User (user roams across devices), Per Device (any user from one device)

Get-RDSessionHost -CollectionName
'Desktop'

List RD Session Hosts in collection

Get-RDUserSession

Show all active RDS user sessions

Invoke-RDUserLogoff -HostServer rdsh1
-UnifiedSessionID 3

Log off specific RDS session

Get-RDLicenseConfiguration

Check RDS licensing configuration

mstsc /v:rdgateway.corp.com
/g:rdgateway.corp.com

RDP via RD Gateway

Section 7: Windows Server Security

Hardening, LAPS, Credential Guard, JEA, Auditing & Tiered Admin

Security Hardening & Baselines

- Server Core: install default — no GUI = smaller attack surface, fewer patches, lower management overhead
- Microsoft Security Compliance Toolkit (SCT): free GPO baselines — Policy Analyzer + LGPO tool for applying baselines
- Disable SMBv1: critical — vulnerability exploited by WannaCry/NotPetya. Disable on ALL servers and clients
- LAPS (Local Administrator Password Solution): unique random local admin passwords per machine stored in AD — prevents lateral movement
- Tiered Administration: Tier 0 (DCs/AD) □ Tier 1 (Servers) □ Tier 2 (Workstations). Admin credentials NEVER cross tiers downward
- PAW (Privileged Access Workstation): dedicated hardened workstation for admin tasks only — no email/browsing

Set-SmbServerConfiguration

-EnableSMB1Protocol \$false

Disable SMBv1

Get-ADComputer -Filter * -Properties

ms-Mcs-AdmPwd | Select Name,

'ms-Mcs-AdmPwd'

Retrieve LAPS passwords

Find-AdmPwdExtendedRights -Identity

OU=Servers,DC=corp,DC=com

Find who can read LAPS passwords

Set-AdmPwdComputerSelfPermission

-Identity OU=Servers,DC=corp,DC=com

Grant computers permission to update own LAPS password

Get-MpComputerStatus | Select

AMRunningMode,

RealTimeProtectionEnabled

Check Windows Defender status

Credential Protection

- Credential Guard: uses VBS (Virtualization-Based Security) to isolate LSASS — prevents pass-the-hash/pass-the-ticket
- Credential Guard requirements: UEFI 2.3.1+, Secure Boot, 64-bit CPU with virtualization, IOMMU (strongly recommended)
- Protected Users group: Kerberos AES-only, no NTLM, no delegation, no credential caching, TGT capped at 4 hours
- JEA (Just Enough Administration): constrained PowerShell runspace — users run specific cmdlets via virtual admin account
- JEA files: Role Capability (.psrc) = allowed commands. Session Configuration (.pssc) = who gets which roles
- All JEA sessions logged to transcript files — full audit trail of every command run

Get-ComputerInfo Select DeviceGuardSmartStatus, DeviceGuardSecurityServicesRunning	Check Credential Guard / Device Guard status
Register-PSSessionConfiguration -Path .\JEAConfig.pssc -Name JEAEndpoint	Register JEA endpoint
Enter-PSSession -ComputerName server1 -ConfigurationName JEAEndpoint	Connect to JEA endpoint
Get-PSSessionConfiguration	List all registered PS session configurations
Add-ADGroupMember -Identity 'Protected Users' -Members 'DomainAdmin1'	Add user to Protected Users

Auditing & Event Monitoring

- Enable Advanced Audit Policy (Computer Configuration □ Security Settings □ Advanced Audit Policy) — more granular than Basic
- Critical event IDs: 4624 (logon), 4625 (failed logon), 4740 (account lockout), 4776 (NTLM), 4768/4769 (Kerberos TGT/ST)
- 4688 (process creation) + command-line logging: enable to detect malicious PowerShell/cmd usage
- 1102 (Security log cleared) + 4719 (audit policy changed) + 7045 (service installed) = high-priority alerts
- Object Access auditing requires SACL on resource object + audit policy enabled — expensive, use selectively

AuditPol /get /category:*	Show current advanced audit policy settings
AuditPol /set /subcategory:'Logon' /success:enable /failure:enable	Enable logon auditing
Get-WinEvent -LogName Security -FilterHashtable @{Id=4625} -MaxEvents 50	Get last 50 failed logons
Get-WinEvent -LogName Security -FilterHashtable @{Id=4740}	Find all account lockout events
wevtutil qe Security /q:"*[System[EventID=4625]]" /c:10 /f:text	Query events with wevtutil

Section 8: High Availability & Clustering

WSFC, Quorum, SQL AG, Storage Replica & Azure DR

Windows Server Failover Clustering

- WSFC: group of servers sharing storage — automatic failover when node fails. Min 2 nodes, max varies by version
- Quorum: prevents split-brain. Types: Node Majority (odd nodes), Node+Disk Witness, Node+File Share Witness, Cloud Witness
- Cloud Witness: Azure Blob Storage as quorum witness — ideal for geo-distributed clusters. Requires Windows Server 2016+
- CSV (Cluster Shared Volumes): all nodes read/write same NTFS volume simultaneously — required for Hyper-V Live Migration + S2D
- CAU (Cluster-Aware Updating): orchestrates node-by-node patching — drains VMs, patches, reboots, resumes. Maintains app availability
- Run validation before creating or changing cluster: Test-Cluster — Microsoft only supports validated clusters

Test-Cluster -Node server1,server2 -Include 'Storage','Network'	Validate cluster prerequisites
New-Cluster -Name Cluster1 -Node server1,server2 -StaticAddress 10.0.0.10	Create failover cluster
Set-ClusterQuorum -CloudWitness -AccountName mystorageacct -AccessKey 'key=='	Configure cloud witness
Suspend-ClusterNode -Name Node1 -Drain -Wait	Drain VMs before maintenance
Resume-ClusterNode -Name Node1 -Failback Immediate	Return node to service after maintenance
Get-ClusterNode Select Name, State, DrainStatus	Check all cluster node states

SQL Server HA — FCI vs Availability Groups

- FCI (Failover Cluster Instance): single SQL instance on shared storage — active/passive. Instance-level HA
- Always On AG: separate SQL instances with database-level log-stream replication. Readable secondaries. No shared storage
- AG synchronous commit: RPO=0 (no data loss), automatic failover. Async commit: low latency, potential data loss
- AG Listener: virtual name/IP clients connect to — routes to primary automatically after failover
- RTO difference: FCI ~5-10 min (WSFC failover + SQL startup). AG ~30 sec (automatic with synchronous commit)

- Choose FCI for: single instance with full SQL feature set. Choose AG for: read scale, multi-site DR, near-zero RTO/RPO

Get-ClusterGroup Where Name -like 'SQL*'	List SQL cluster roles
Move-ClusterGroup -Name 'SQL Server (MSSQLSERVER)' -Node Node2	Manual SQL FCI failover
SELECT * FROM sys.dm_hadr_availability_group_states	Check AG health in SQL
ALTER AVAILABILITY GROUP [AG1] FAILOVER	Manual AG failover (T-SQL)
Get-ClusterResource Where Name -like '*Listener*'	Find AG listener cluster resource

Storage Replica & Disaster Recovery

- Storage Replica (Datacenter): block-level replication. Synchronous = RPO 0. Async = lower latency, possible loss
- Destination volume is offline/read-only during replication — cannot be accessed until failover
- RTO vs RPO: RTO = how long to restore service. RPO = how much data can be lost. Synchronous replication = RPO 0
- Azure Site Recovery (ASR): replicate Hyper-V/VMware/physical servers to Azure. RTO 15-30 min. Supports DR drills (non-disruptive)
- Stretched cluster: WSFC across two sites with synchronous Storage Replica — automatic failover between datacenters

New-SRPartnership -SourceComputerName SRV1 -SourceRGName SRG1 -SourceVolumeName D: -DestinationComputerName SRV2 -DestinationRGName DRG1 -DestinationVolumeName D: -ReplicationMode Synchronous	Create synchronous Storage Replica
Get-SRPartnership	Show Storage Replica partnership status
Get-SRGroup Get-SRPartnership Select ReplicationStatus, LastSyncTime	Check replication health
(Get-SRGroup).Replicas.ReplicationStatus	Detailed replica status
Set-SRPartnership -NewSourceComputerName SRV2 -SourceRGName DRG1 -DestinationComputerName SRV1 -DestinationRGName SRG1	Failover Storage Replica (swap direction)

Section 9: Monitoring & Troubleshooting

Performance, Event Logs, SFC/DISM, AD Diagnostics & Sysinternals

Performance Monitoring

- perfmon.exe: Performance Monitor — real-time counters, data collector sets, performance logs
- Key counters: Processor\% Processor Time (CPU), Memory\Available MBytes (RAM), PhysicalDisk\Avg Disk Queue Length (I/O), Network\Bytes Total/sec
- resmon.exe (Resource Monitor): real-time CPU/memory/disk/network per process — faster for quick diagnosis
- Disk Queue Length > 2 per disk = I/O bottleneck. High iowait% in Task Manager CPU section = disk bottleneck
- WinSAT: winsat formal — baseline hardware capability scores. Reliability Monitor: stability index + failure history

Get-Counter -Counter '\Processor(_Total)\% Processor Time' -SampleInterval 2 -MaxSamples 5	Sample CPU counter 5 times
Get-Counter -ListSet Memory Select -Expand Counter	List all available Memory counters
typeperf '\PhysicalDisk(_Total)\Avg. Disk Queue Length' -si 1 -sc 10	Monitor disk queue from command line
logman create counter PerfLog -c '\Processor\% Processor Time' -si 5 -o C:\perflog.blg	Create performance data collector
winsat formal	Run Windows System Assessment Tool

Event Logs & Troubleshooting Tools

- Event Viewer key logs: System (services/drivers), Application, Security (audit), Directory Service (AD), DNS Server, DFS Replication
- SFC /scannow: scan and repair system files. Run DISM /RestoreHealth FIRST to repair component store that SFC uses
- Process Monitor (ProcMon): Sysinternals — captures file/registry/network/process activity. Filter by process + result=ACCESS DENIED
- Process Explorer: Sysinternals — detailed process tree, parent-child relationships, loaded DLLs, security context
- Autoruns: Sysinternals — all auto-starting programs (startup, services, drivers, scheduled tasks) — find malware persistence
- Boot issues: run bootrec /fixmbr, /fixboot, /rebuildbcd from Windows RE. SFC offline for boot file repair

DISM /Online /Cleanup-Image /RestoreHealth	Repair Windows component store (run BEFORE SFC)
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sfc /scannow	Scan and repair corrupted Windows system files
Get-WinEvent -LogName System -FilterHashtable @{Level=2} -MaxEvents 20	Get last 20 System errors
wevtutil qe Application /c:50 /f:text findstr 'Error'	Query Application log for errors
netsh winsock reset && netsh int ip reset	Reset TCP/IP and Winsock stack
bcdedit /set {default} safeboot minimal	Configure next boot into Safe Mode

Active Directory Diagnostics

- dcdiag: comprehensive DC health tests — run dcdiag /e for entire forest, dcdiag /test:dns for DNS only
- repadmin: AD replication health — /showrepl (status), /replsummary (overview), /syncall (force sync)
- Common replication errors: 8453 (access denied), 1722 (RPC unavailable — check DNS/firewall), 8457 (replication disabled)
- Check: DNS (nslookup _ldap._tcp), time sync (w32tm /query /status), firewalls (ports 135, 49152+, 389, 88, 445)
- netlogon.log: enable with nltest /dbflag:0x2080ffff — diagnoses logon/authentication failures in detail

dcdiag /s:dc1 /test:replications /test:dns /test:netlogons	Run key DC health tests
repadmin /showrepl * /errorsonly	Show only DCs with replication errors
dcdiag /e /test:dns > dns-report.txt	Full forest DNS test to file
nltest /dbflag:0x2080ffff	Enable verbose NETLOGON logging
nltest /dbflag:0x0	Disable verbose NETLOGON logging
Test-ComputerSecureChannel -Repair	Repair broken domain trust (secure channel)

Section 10: Hybrid & Cloud Integration

Azure Arc, Azure Backup, File Sync, WAC & Exam Quick Reference

Azure Hybrid Services

- Azure Arc: enroll on-prem servers in Azure — apply Azure Policy, Monitor, Defender, Update Management, RBAC
- Azure Backup (MARS agent): back up files/folders/system state from any Windows Server to Recovery Services Vault
- Azure File Sync: Windows Server becomes intelligent cache for Azure Files. Cloud tiering = rarely-used files in cloud, hot files local
- Azure Update Management: patch on-prem + Azure VMs from one console — no WSUS needed
- Azure AD Application Proxy: publish internal web apps externally via Azure AD — no VPN, no WAP. Connector = outbound HTTPS only
- Azure AD Connect: sync on-prem AD to Azure AD. Methods: PHS (hash sync), PTA (pass-through auth), Federation (ADFS)

azcmagent connect --resource-group RG1 --tenant-id --location eastus --subscription-id	Enroll server with Azure Arc
azcmagent show	Show Azure Arc agent status and configuration
Start-MARSAgentJob	Trigger on-demand Azure Backup from MARS agent
Register-AzStorageSyncServer -ResourceGroupName RG1 -StorageSyncServiceName SyncSvc	Register server with Azure File Sync
Connect-AzureAD && Get-AzureADUser	Connect to Azure AD via PowerShell

Windows Server on Azure

- Azure Hybrid Benefit: use existing Windows Server licenses with SA — pay compute only, save 40-80% on Azure VMs
- Azure Automanage: auto-enables Backup, Monitor, Defender, Update Mgmt, Change Tracking with one click
- Windows Server 2022 Azure Edition: Hotpatch (security patches without reboot), SMB over QUIC, Trusted Launch
- Hotpatch: apply security updates monthly without reboots between quarterly baseline updates — reduces reboots by 60%
- Secured-core Server: hardware-backed security — TPM 2.0, Secure Boot, DRTM, VBS, HVCI. Requires OEM-certified hardware

Set-AzVm -ResourceGroupName RG1 -Name VM1 -LicenseType Windows_Server	Apply Azure Hybrid Benefit to existing VM
Get-AzVM -ResourceGroupName RG1 -Name VM1 Select -Expand LicenseType	Check if Hybrid Benefit is applied

Set-AzVMPatchAssessmentMode -ResourceGroupName RG1 -VMName VM1 -Mode AutomaticByPlatform	Enable automatic patch assessment
Get-WindowsUpdateLog	Get Windows Update history log
Get-ComputerInfo Select SecureBoot, VirtualizationFirmwareEnabled, HyperVisorPresent	Check security baseline features

AZ-800 Exam Quick Reference

- Exam domains: Deploy and manage AD DS (30-35%) | Manage Windows Server (15-20%) | Manage VMs and containers (15-20%) | Implement and manage storage (15-20%) | Implement and manage network infrastructure (10-15%)
- Key tools to know: dcdiag, repadmin, netdom, nltest, ntdsutil, gpresult, gpupdate, w32tm, setspn, nslookup, ipconfig
- FSMO exam trap: know which roles are forest-wide (Schema, Domain Naming) vs domain-wide (PDC, RID, Infrastructure)
- Replication exam tip: intra-site = immediate, inter-site = scheduled. KCC builds topology automatically
- DHCP failover modes: Hot Standby (one active, one standby with 5% reserve) vs Load Sharing (both active, split addresses)
- Storage exam tip: ReFS = integrity + block cloning for Hyper-V. NTFS = compression + EFS + quotas for file servers

dcdiag /e /c	Run ALL tests on ALL DCs in forest – comprehensive health check
repadmin /showrepl * /csv > repl.csv	Export replication status to CSV for reporting
netdom trust DOMAIN1 /d:DOMAIN2 /verify	Verify trust between domains
Get-ADComputer -Filter * -Properties LastLogonDate Where LastLogonDate -lt (Get-Date).AddDays(-90)	Find computers not logged in for 90+ days
Get-ADUser -Filter {Enabled -eq \$true -and PasswordNeverExpires -eq \$true}	Find users with password never expires
Test-Cluster -Node node1,node2 -Include 'Storage','Network','System Configuration'	Comprehensive cluster validation